CO 5420 - Project Specification

Objectives

By the end of this project, you should be able to :

a. **Formulate a question in Deep Learning and justify the solution**
b. **Implement a solution including a variant of Artificial Neural Network**
c. **Self-assess your contribution to a project**

1. Project Title Selection

In the CO5420 - Neural Networks and Deep Learning project, our main focus is to give you the hands-on experience of the things we are going to cover in the course, as well as giving you the chance to explore new things on your own in this field. Consider this project a 6-week mini project (including the proposal planning) on DL.

AI level of use: Level 3 – AI Collaboration AI may be used to help complete the task, including idea generation, drafting, feedback, and refinement. You should critically evaluate and modify the AI suggested outputs, demonstrating your understanding.

The set of problems for the project are here.

Each project consists of a kaggle task and an extended task; Kaggle task requires you to develop an appropriate model to the given problem and it will be tested on an unseen dataset and ranked across a group of teams. See more about Kaggle competitions here.

For the extended task you can take either a hands-on approach or research approach for the extended task. Example handson would be where you would be applying a range of existing methods whereas example research approach would be you would be extending on a key research idea related to your topic.

2. Project Groups

Each project group would have 5 members per group. You are expected to work on one selected project of your interest as a group. **Use this Google Form to add your group members by EoD 31.05.2026 .** In order to balance the class and number of groups we may add one or two members to your group when finalising the groups.

3. Project Deliverables

A. Project Proposal (10%) (Scheduled by Week 7)

After finalizing your project topics, we would like to sit with your group and plan the project during week 7. Your participation in this activity is compulsory.

Work as a group and submit your plan for the project in the prescribed format here by EoD **10.06.2026** through LMS as a PDF named as EXX_YYY.pdf where XX is your batch and YYY is your group No**. here** (if there are members from both E/22 and E/23 then, XX is 22_23)

This is mainly done to evaluate your project ideas and let you recalibrate your ideas before you start to work on your project.

B. End Evaluation (90%) (Scheduled for Week 13)

The end evaluation consists of three components: your code log in Kaggle (20%), model performance and ranking in Kaggle (10%), final presentation and an individual viva (60%).

A randomly set out sample from the dataset will be provided to you in Week 12. You have to submit your predictions to the mentioned dataset. The performances of each group submission will be calculated and the groups will be ranked based on performance.

Final presentation and individual viva

(Poster)Presentation:
In this you will present a summary of all your work as a group. It can be presented by one/few/all group members.

The objective of a poster presentation is to enable someone outside your group to understand your work without you presenting it. Read more about poster presentations [here.](#)

Your poster must include:

1. Title, with your names and E/Nos
2. Abstract: A paragraph or two summarizing your work.
3. Introduction: Provide a background to your problem and why it is significant
4. Problem: Provide details about the problem: How data look like, what exactly is the problem solved?
5. Methodology: Provide details of the approaches you have taken to infer from the data.
6. Results: Present key results of the work
7. Discussion and conclusions: What are the implications? What conclusions you can make based on your findings. Any limitations of your work, what would be possible future work?
8. References: Provide the list of references
9. Declaration on the use of Gen AI tools.

Find a poster template [here](#). Feel free to modify/innovate.

Component	**Weight**	**What is assessed**
Timing	5%	Presentation is completed within the 3-minute limit.
Methodology	10%	The approach taken, architecture, and justification of design choices.
Analysis	5%	Quality of results interpretation, evaluation metrics, and insights drawn.

Individual viva:

Assessed on: understanding of the methodology, ability to justify decisions made, and awareness of your own contribution to the project.

As the class size this year is large, we will run strictly on the timing. •

If you do not know the answer, say so briefly rather than using up time : do not guess at length.

Datasets and codebases

We have provided a set of standard problems with varying levels of difficulty. You may choose any problem from the list. For easier problems, you will be expected to put in more effort to earn higher marks. This approach is intended to ensure fairness in the assessment process, as all problems will be evaluated equally.

For Any Queries

Contact:

Course Coordinator : Dr. Damayanthi Herath damayanthiherath@eng.pdn.ac.lk

Template for Project Proposal

CO 5420 — Project Draft Plan

Group Details

Motivation/Problem Definition

Dataset Description

Proposed Approach (Initial Ideas)

List 1-2 candidate algorithms/architectures you intend to try first, and your initial idea for how you will evaluate results (metric(s) and any baseline).

- Candidate architecture(s)/algorithm(s):
- Evaluation metric(s) and baseline etc…

Proposed Contribution (Division of Work)

Member Name	Planned Contribution / Task	Approx. Timeline

Generative AI Usage Statement

Fill in the blanks as appropriate

Declaration: by submitting this Draft Plan, the group confirms that the content reflects the group's own understanding and decisions, and that any generative AI assistance has been disclosed above in accordance with the module's plagiarism policy.

References

A few useful resources

You can use department servers to run NNs effectively: https://cepdnaclk.github.io/sites/servers

You can forward the Jupyter notebooks easily:
https://cepdnaclk.github.io/sites/servers/tesla/#how-can-i-run-a-jupyter-notebook-on-the-server-and-access-it-over-the-internet

We have hosted many popular datasets on our servers. You can use them without downloading them again:
https://faq.ce.pdn.ac.lk/network-n-servers/kepler/

Submit the project poster using this link.

Find the final project presentation schedule here.